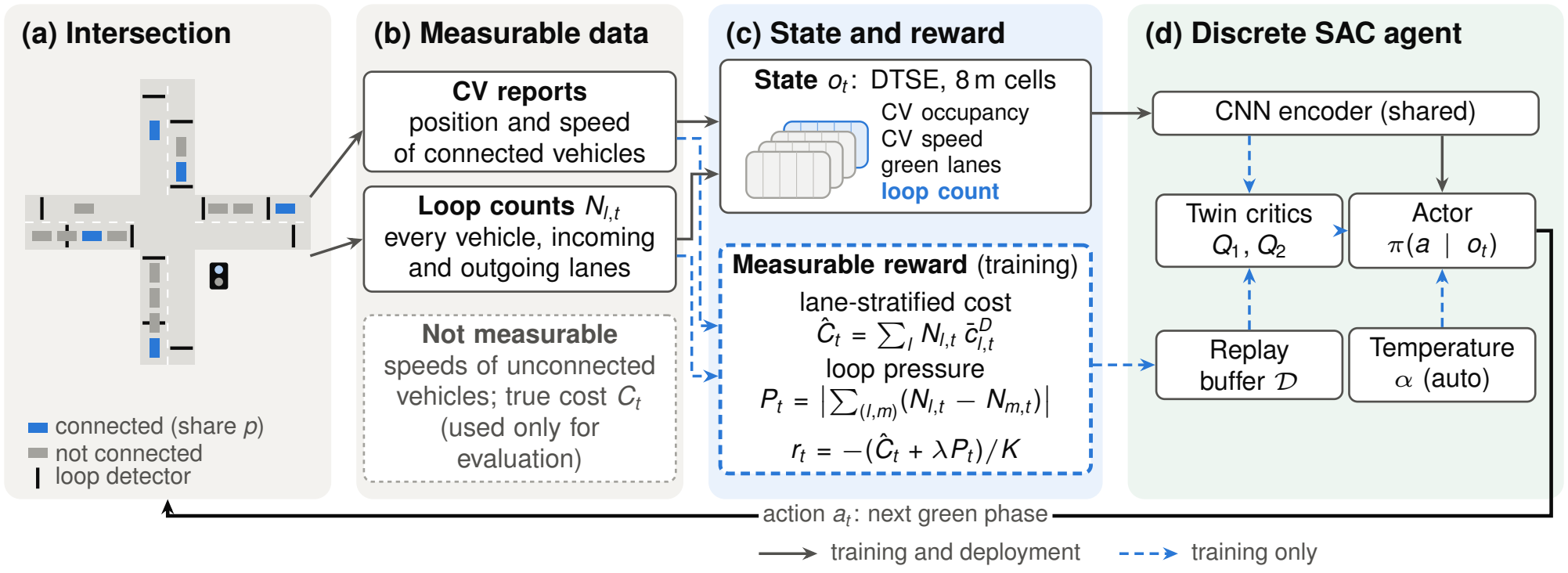
(a) Intersection
connected (share p)
not connected
loop detector
(b) Measurable data
CV reports
position and speed
of connected vehicles
Loop counts $N_{l,t}$
every vehicle, incoming
and outgoing lanes
Not measurable
speeds of unconnected
vehicles; true cost C_t
(used only for
evaluation)
(c) State and reward
State o_t: DTSE, 8 m cells
CV occupancy
CV speed
green lanes
loop count
Measurable reward (training)
lane-stratified cost
$\hat{C}_t = \sum_l N_{l,t}\, \bar{c}^D_{l,t}$
loop pressure
$P_t = \left|\sum_{(l,m)} (N_{l,t} - N_{m,t})\right|$
$r_t = -(\hat{C}_t + \lambda P_t)/K$
(d) Discrete SAC agent
CNN encoder (shared)
Twin critics
Q_1, Q_2
Actor
$\pi(a \mid o_t)$
Replay
buffer $\mathcal{D}$
Temperature
α (auto)
action a_t: next green phase
training and deployment
training only